Glycolysis:
The Central Pathway of Glucose Degradation

BDS SEMESTER II

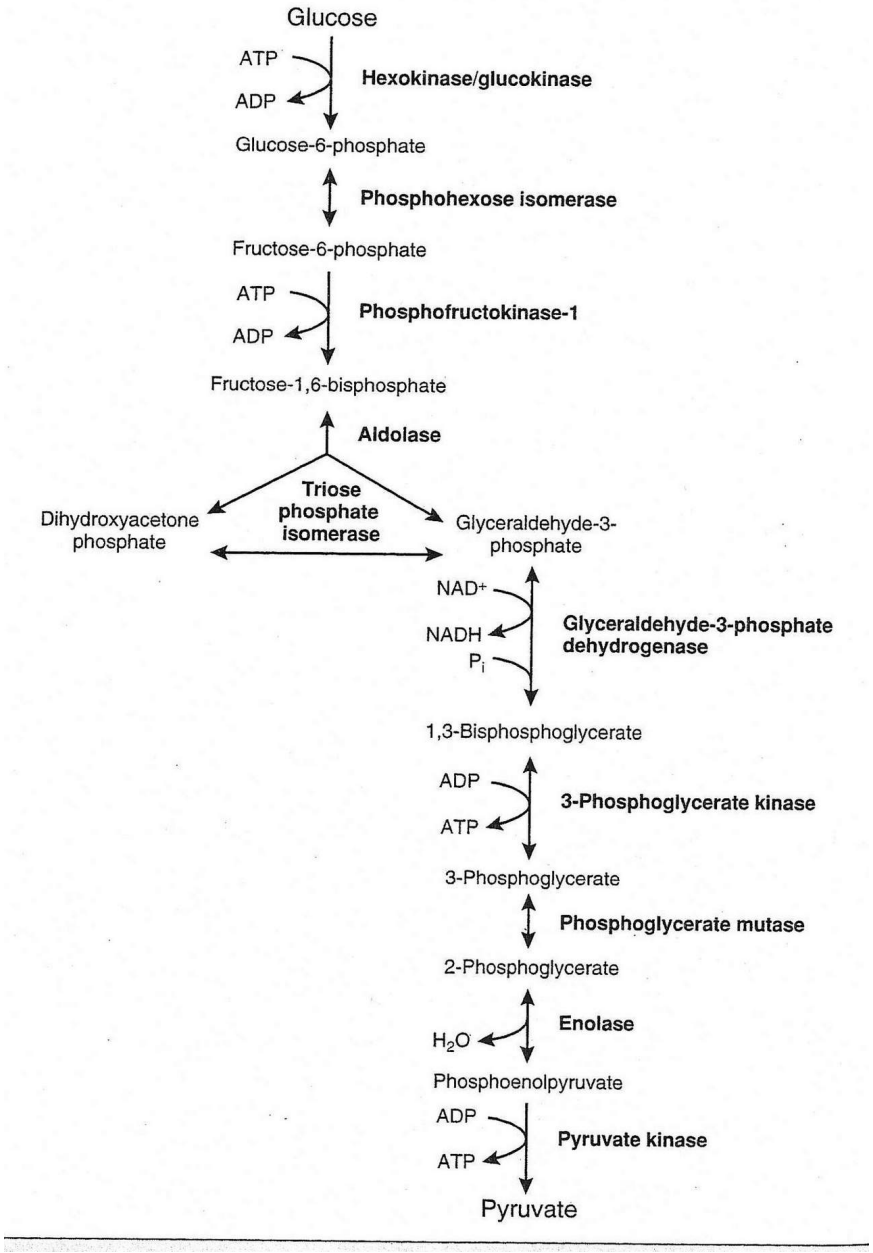
PREPARED BY FATIMA NOOR

Clinical Case:

- 15 y.o. female
  - Hemolytic anemia diagnosed at age 3 mo.
  - Recurrent episodes of pallor, jaundice, leg ulcer
    - Enlarged spleen, low Hb, low RBC count, elevated reticulocyte count
    - Abnormal RBC shape, short RBC life, elevated total and indirect bilirubin
    - RBC with elevated 2,3-BPG and low ATP
  - Following spleenectomy clinical and hematological symptoms improved.

Glycolysis: Embden-Myerhof Pathway

- Oxidation of glucose
- Products:
  - 2 Pyruvate
  - 2 ATP
  - 2 NADH
- Cytosolic

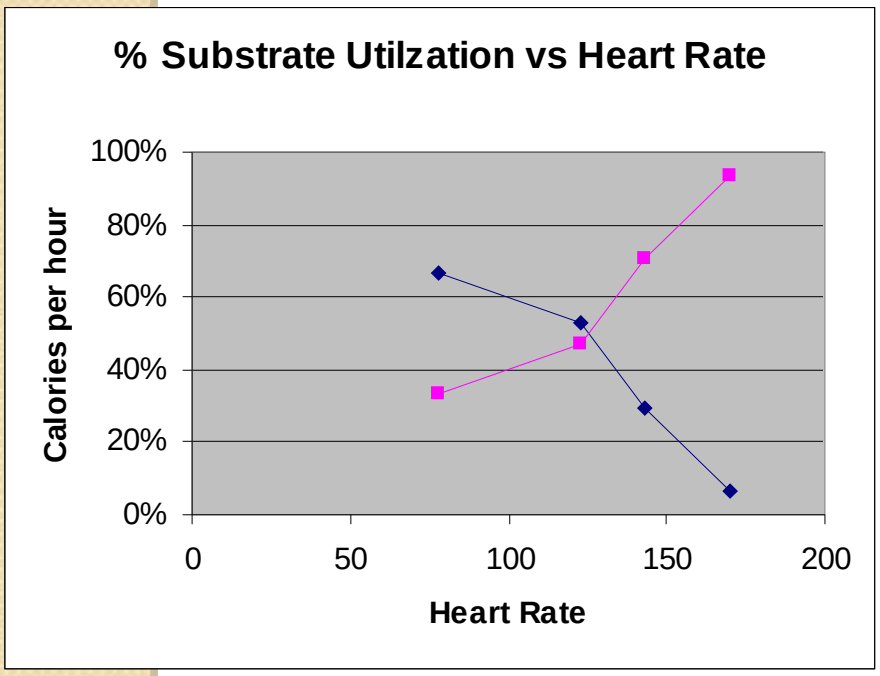


Glycolysis: General Functions

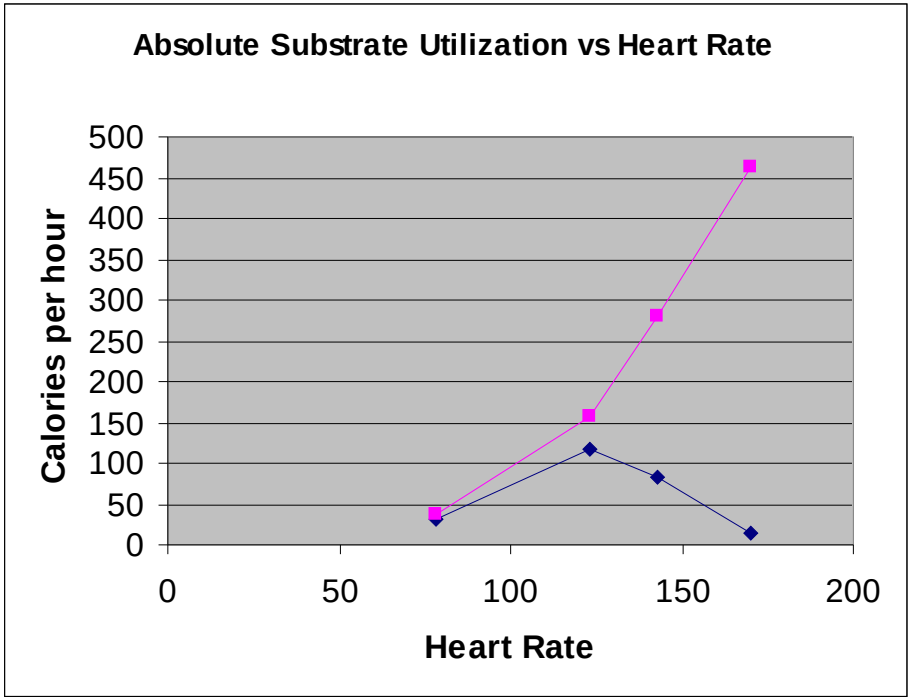
- Provide ATP energy
- Generate intermediates for other pathways
  - Hexose monophosphate pathway
  - Glycogen synthesis
  - Pyruvate dehydrogenase
    - Fatty acid synthesis
    - Krebs' Cycle
  - Glycerol-phosphate (TG synthesis)

Glycolysis: Specific tissue functions

- RBC's
  - Rely exclusively for energy
- Skeletal muscle
  - Source of energy during exercise, particularly high intensity exercise
- Adipose tissue
  - Source of glycerol-P for TG synthesis
  - Source of acetyl-CoA for FA synthesis
- Liver
  - Source of acetyl-CoA for FA synthesis
  - Source of glycerol-P for TG synthesis



Data from 2007 NUTR 442
Indirect Calorimetry Laboratory



Regulation of Cellular Glucose Uptake

- Brain & RBC:
  - GLUT-1 has high affinity (low Km)for glucose and are always saturated.
    - Insures that brain and RBC always have glucose.
- Liver:
  - GLUT-2 has low affinity (hi Km) and high capacity.
    - Uses glucose when fed at rate proportional to glucose concentration
- Muscle & Adipose:
  - GLUT-4 is sensitive to insulin

Glucose Utilization

- Phosphorylation of glucose
  - Commits glucose for use by that cell
  - Energy consuming
- Hexokinase: muscle and other tissues
- Glucokinase: liver

Properties of Glucokinase and Hexokinase

Table 11-1

	Glucokinase	*Hexokinase*
Kinetic parameters		
K_m	High (10 mM)	Low (<100 μM)
V_{max}	High	Low
Tissue distribution	Liver Pancreatic beta cells	Most tissues
Regulation		
Short-term	Activity responds to changes in glucose concentration	Inhibited by glucose-6-phosphate
Long-term	Synthesis induced by insulin	Constitutive

Regulation of Cellular Glucose Utilization in the Liver

- **Feeding**
  - Blood glucose concentration high
  - GLUT-2 taking up glucose
  - Glucokinase induced by insulin
  - High cell glucose allows GK to phosphorylate glucose for use by liver
- **Post-absorptive state**
  - Blood & cell glucose low
  - GLUT-2 not taking up glucose
  - Glucokinase not phophorylating glucose
  - Liver not utilizing glucose during post-absorptive state

Regulation of Cellular Glucose Utilization in the Liver

- Starvation
  - Blood & cell glucose concentration low
  - GLUT-2 not taking up glucose
  - GK synthesis repressed
  - Glucose not used by liver during starvation

Regulation of Cellular Glucose Utilization in the Muscle

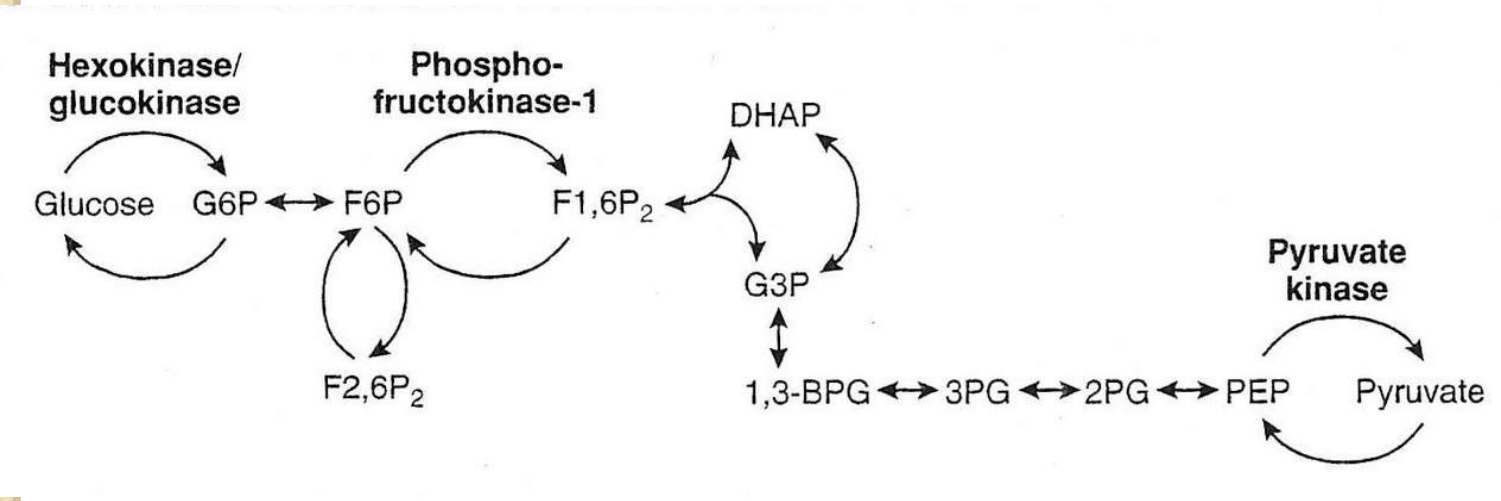
- Feeding and at rest
  - High blood glucose, high insulin
  - GLUT-4 taking up glucose
  - HK phosphorylating glucose
  - If glycogen stores are filled, high G6P inhibits HK, decreasing glucose utilization
- Starving and at rest
  - Low blood glucose, low insulin
  - GLUT-4 activity low
  - HK constitutive
  - If glycogen stores are filled, high G6P inhibits HK, decreasing glucose utilization

Regulation of Cellular Glucose Utilization in the Muscle

- Exercising Muscle (fed or starved)
  - Low G6P (being used in glycolysis)
  - No inhibition of HK
  - High glycolysis from glycogen or blood glucose

Regulation of Glycolysis

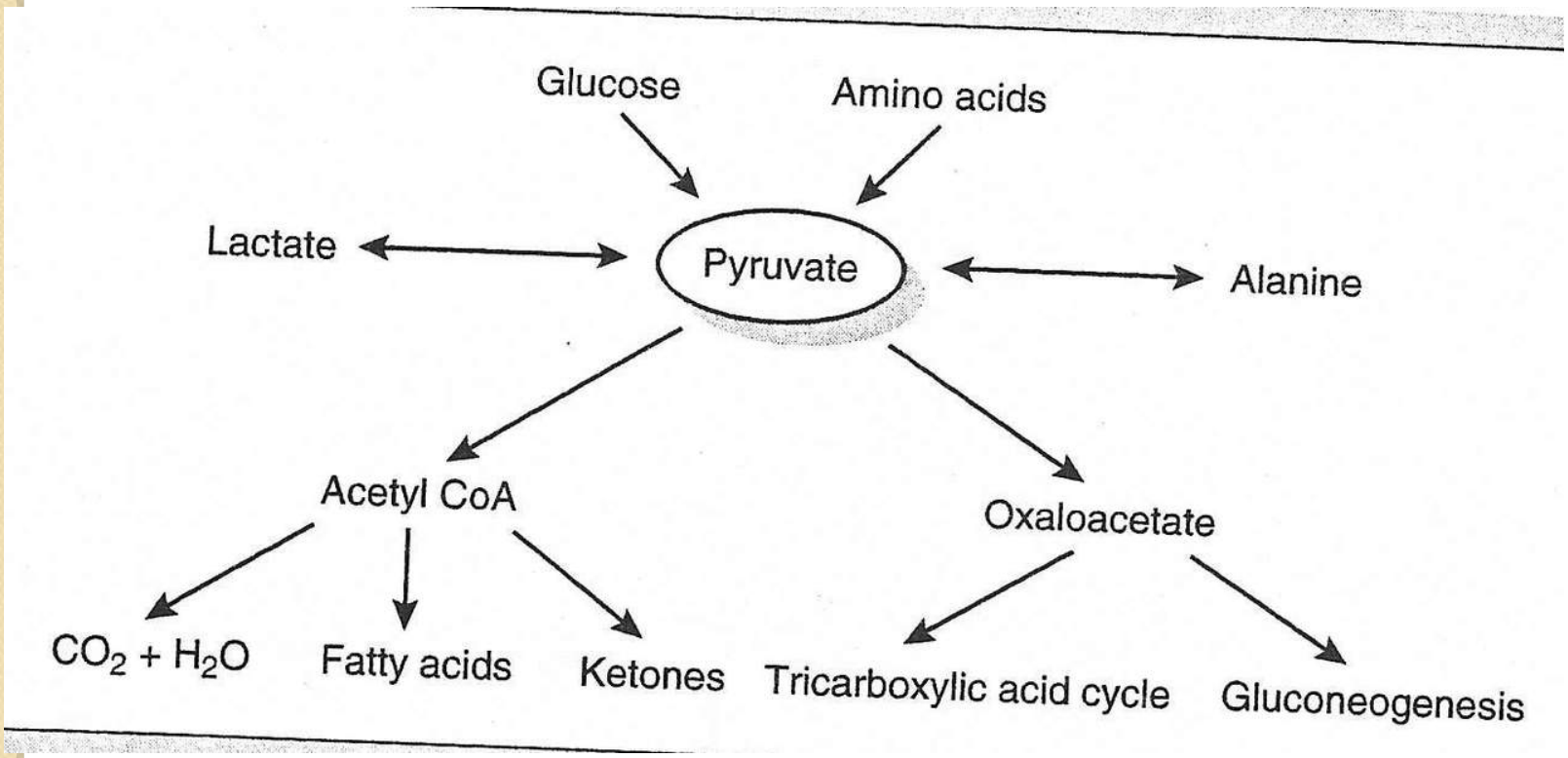
- Regulation of 3 irreversible steps
- PFK-1 is rate limiting enzyme and primary site of regulation.



Pyruvate Dehydrogenase:

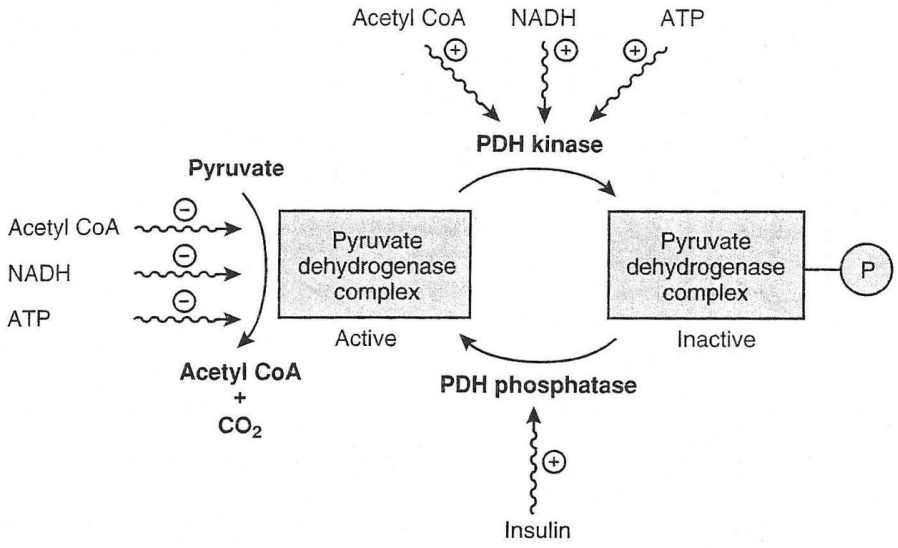
The enzyme that links glycolysis with other pathways

- Pyruvate + CoA + NAD → AcetylCoA + CO2 + NADH



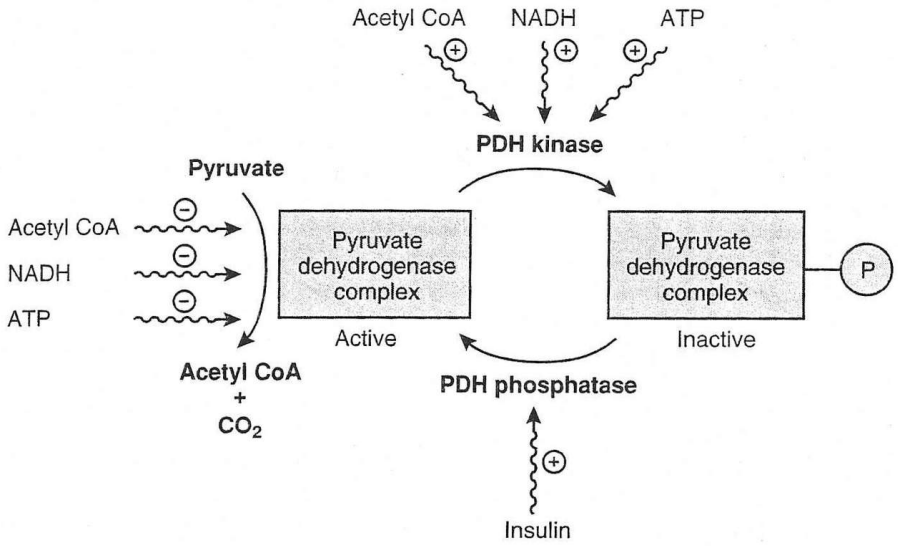
Regulation of PDH
Muscle

- Resting (don't need)
  - Hi energy state
  - Hi NADH & AcCoA
    - Inactivates PDH
  - Hi ATP & NADH & AcCoA
    - Inhibits PDH
- Exercising (need)
  - Low NADH, ATP, AcCoA



Regulation of PDH
Liver

- Fed (need to make FA)
  - Hi energy
  - Insulin activates PDH
- Starved (don't need)
  - Hi energy
  - No insulin
    - PDH inactive



Clinical Case:
Pyruvate Kinase Deficiency

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Clinical Case:
Pyruvate Kinase Deficiency

- RBC dependent on glycolysis for energy
  - Sodium/potassium ion pumps require ATP
  - Abnormal RBC shape a result of inadequate ion pumping
    - Excessive RBC destruction in spleen
      - Hemolysis
      - Jaundice (elevated bilirubin, fecal urobilinogens)
      - Increased reticulocyte count

Clinical Case:

Pyruvate Kinase Deficiency

- <10% activity of PK
  - Results in increase in glycolytic intermediates (2,3-BPG)
  - Recessive autosomal disorders of isozyme found only in RBC's
  - Heterozygous defect occurs in about 1% of Americans
    - Second most common genetic cause of hemolytic anemia (G6PDH deficiency #1)
    - Rare (51/million Caucasian births, may be underdiagnosed)

